Regulatory Capture via Relative Prices

Gabriel Cañedo Riedel
UC San Diego

October 2026

The Pattern

- **Grocery Manufacturers of America (GMA):** lobbied for the Food Safety Modernization Act (FSMA, 2016), the biggest reform of U.S. food-safety law since 1938.
- **Amazon** asked Congress to double the federal minimum wage.
- **Apple** asked Congress for stricter data-privacy law.
- **Meta** asked Congress for content-moderation legislation.

The Pattern

- **Grocery Manufacturers of America (GMA):** lobbied for the Food Safety Modernization Act (FSMA, 2016), the biggest reform of U.S. food-safety law since 1938.
- **Amazon** asked Congress to double the federal minimum wage.
- **Apple** asked Congress for stricter data-privacy law.
- **Meta** asked Congress for content-moderation legislation.

Why would firms ask for regulations that increase their own costs?

Proposed Mechanism

If a regulation raises some firms' marginal costs proportionally more than others', some firms profit:

Proposed Mechanism

If a regulation raises some firms' marginal costs proportionally more than others', some firms profit:

- Every price rises, but the least burdened firms raise theirs the **least**.
- They become **relatively** cheaper, attract demand, and grow: they gain revenue, profit, and market share.

Proposed Mechanism

If a regulation raises some firms' marginal costs proportionally more than others', some firms profit:

- Every price rises, but the least burdened firms raise theirs the **least**.
- They become **relatively** cheaper, attract demand, and grow: they gain revenue, profit, and market share.

Focus: regulations with **regressive incidence**, that is, the increase in marginal cost decreases with productivity.

The Pattern

- **GMA:** the largest food manufacturers lobbied for food-safety law.
  It largely put their existing practices into law; for smaller manufacturers, it meant new programs and personnel.

The Pattern

- **GMA:** the largest food manufacturers lobbied for food-safety law.
  It largely put their existing practices into law; for smaller manufacturers, it meant new programs and personnel.
- **Amazon:** double the federal minimum wage.
  It already pays above the proposed minimum, so it is exposed only indirectly, through suppliers and outsourced services.

The Pattern

- **GMA:** the largest food manufacturers lobbied for food-safety law.
  It largely put their existing practices into law; for smaller manufacturers, it meant new programs and personnel.
- **Amazon:** double the federal minimum wage.
  It already pays above the proposed minimum, so it is exposed only indirectly, through suppliers and outsourced services.
- **Apple:** stricter data-privacy law.
  Unlike its competitors, its revenue does not depend on targeted ads.

The Pattern

- **GMA:** the largest food manufacturers lobbied for food-safety law.
  It largely put their existing practices into law; for smaller manufacturers, it meant new programs and personnel.
- **Amazon:** double the federal minimum wage.
  It already pays above the proposed minimum, so it is exposed only indirectly, through suppliers and outsourced services.
- **Apple:** stricter data-privacy law.
  Unlike its competitors, its revenue does not depend on targeted ads.
- **Meta:** content-moderation legislation.
  It already runs the largest content-moderation infrastructure; smaller platforms would have to build one.

Why regressive incidence matters

Regressive incidence makes large firms cheaper relative to smaller firms. It reallocates demand toward them, making the size distribution of firms more unequal.

Why regressive incidence matters

Regressive incidence makes large firms cheaper relative to smaller firms. It reallocates demand toward them, making the size distribution of firms more unequal.

The firm-size distribution has grown more unequal for at least a century (Kwon et al., 2024; Covarrubias et al., 2020). Driving:

- Labor's declining share (Autor et al., 2020).
- The rise in the aggregate markup (De Loecker et al., 2020), which lowers productivity and welfare (Edmond et al., 2023).
- Aggregate instability (Gabaix, 2011).

Why regressive incidence matters

Regressive incidence makes large firms cheaper relative to smaller firms. It reallocates demand toward them, making the size distribution of firms more unequal.

The firm-size distribution has grown more unequal for at least a century (Kwon et al., 2024; Covarrubias et al., 2020). Driving:

- Labor's declining share (Autor et al., 2020).
- The rise in the aggregate markup (De Loecker et al., 2020), which lowers productivity and welfare (Edmond et al., 2023).
- Aggregate instability (Gabaix, 2011).

These literatures do not study regulation, but they show why regulations that reshape the firm-size distribution matter.

Literature Review

Demand reallocation through **relative prices** is missing from the literature on cost-increasing regulation.

Literature Review

Demand reallocation through **relative prices** is missing from the literature on cost-increasing regulation.

- Marginal-cost regulations that burden some firms more are studied as *raising rivals' costs*: homogeneous goods, partial equilibrium (Williamson, 1968; Salop and Scheffman, 1983; Fowlie et al., 2016).
  But homogeneous goods have a **single price**.

Literature Review

Demand reallocation through **relative prices** is missing from the literature on cost-increasing regulation.

- Marginal-cost regulations that burden some firms more are studied as *raising rivals' costs*: homogeneous goods, partial equilibrium (Williamson, 1968; Salop and Scheffman, 1983; Fowlie et al., 2016).
  But homogeneous goods have a **single price**.
- In differentiated goods, regulations that affect firms differently are studied only through fixed costs (selection) (Rebeyrol, 2023; Macedoni and Weinberger, 2026).
  But selection leaves the relative prices among survivors **unchanged**.

Literature Review

Demand reallocation through **relative prices** is missing from the literature on cost-increasing regulation.

- Marginal-cost regulations that burden some firms more are studied as *raising rivals' costs*: homogeneous goods, partial equilibrium (Williamson, 1968; Salop and Scheffman, 1983; Fowlie et al., 2016).
  But homogeneous goods have a **single price**.
- In differentiated goods, regulations that affect firms differently are studied only through fixed costs (selection) (Rebeyrol, 2023; Macedoni and Weinberger, 2026).
  But selection leaves the relative prices among survivors **unchanged**.

This paper: *raising rivals' costs* in differentiated goods, brought to general equilibrium.

Outline

- Develop the relative-price mechanism in CES partial equilibrium.
  A firm gains when a regulation raises the price index more than its own marginal cost.
- Derive the estimating equation.
- Apply it to FSMA with Nielsen scanner data.
  Large firms grow relative to small firms in affected products, but not in placebos.
- Bring the mechanism to general equilibrium.
  It strengthens when demand across industries is inelastic.
- Quantify FSMA's welfare cost.
  The cost is at least $362M a year.

Environment: Consumers

Consider a static economy with a unit continuum of differentiated varieties $\omega \in (0, 1)$.

- Consumers have CES preferences with $\sigma > 1$ (gross substitutes). Expenditure Y is fixed.
- Demand for variety ω:

$$x_\omega = \frac{Y}{p_\omega} \left(\frac{p_\omega}{P(\mathbf{p})} \right)^{1-\sigma}$$

  p_ω is the price of variety ω, $\mathbf{p}$ the price profile, and $P(\mathbf{p}) = \left(\int_0^1 p_\omega^{1-\sigma} \, d\omega \right)^{\frac{1}{1-\sigma}}$ the price aggregator.

Environment: Firms in monopolistic competition

- Each variety ω is produced by a single firm with constant marginal cost $\psi_\omega > 0$, which takes P as given and chooses its price to maximize profit:

$$\pi_\omega \;=\; (p_\omega - \psi_\omega)\, x_\omega \;=\; \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y \left(\frac{p_\omega}{P(\mathbf{p})}\right)^{1-\sigma}$$

Environment: Firms in monopolistic competition

- Each variety ω is produced by a single firm with constant marginal cost $\psi_\omega > 0$, which takes P as given and chooses its price to maximize profit:

$$\pi_\omega \;=\; (p_\omega - \psi_\omega)\, x_\omega \;=\; \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y \left(\frac{p_\omega}{P(\mathbf{p})}\right)^{1-\sigma}$$

- Write $z_\omega \equiv p_\omega / P(\mathbf{p})$ for the relative price. The same problem becomes:

$$\pi_\omega \;=\; \left(1 - \frac{\psi_\omega / P(\mathbf{p})}{z_\omega}\right) Y\, z_\omega^{1-\sigma}$$

Environment: Firms in monopolistic competition

- Each variety ω is produced by a single firm with constant marginal cost $\psi_\omega > 0$, which takes P as given and chooses its price to maximize profit:

$$\pi_\omega \;=\; (p_\omega - \psi_\omega)\, x_\omega \;=\; \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y \left(\frac{p_\omega}{P(\mathbf{p})}\right)^{1-\sigma}$$

- Write $z_\omega \equiv p_\omega / P(\mathbf{p})$ for the relative price. The same problem becomes:

$$\pi_\omega \;=\; \left(1 - \frac{\psi_\omega / P(\mathbf{p})}{z_\omega}\right) Y\, z_\omega^{1-\sigma}$$

- Maximizing over price levels is the same as maximizing over relative prices. Marginal cost enters the problem only through **relative marginal cost** ψ_ω / P.

Environment: Firms in monopolistic competition

- Each variety ω is produced by a single firm with constant marginal cost $\psi_\omega > 0$, which takes P as given and chooses its price to maximize profit:

$$\pi_\omega \;=\; (p_\omega - \psi_\omega)\, x_\omega \;=\; \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y \left(\frac{p_\omega}{P(\mathbf{p})}\right)^{1-\sigma}$$

- Write $z_\omega \equiv p_\omega / P(\mathbf{p})$ for the relative price. The same problem becomes:

$$\pi_\omega \;=\; \left(1 - \frac{\psi_\omega / P(\mathbf{p})}{z_\omega}\right) Y\, z_\omega^{1-\sigma}$$

- Maximizing over price levels is the same as maximizing over relative prices. Marginal cost enters the problem only through **relative marginal cost** ψ_ω / P.

Relative marginal cost ψ_ω / P is the sufficient statistic: it determines the firm's equilibrium relative price, market share, revenue, and profit.

Regulation

Index firms by marginal cost: $\psi_\omega = G^{-1}(\omega)$, for a continuous, strictly increasing cost distribution G on $[\psi_0, \infty)$, with $\psi_0 > 0$. The most productive firms sit near $\omega = 0$.

Regulation

Index firms by marginal cost: $\psi_\omega = G^{-1}(\omega)$, for a continuous, strictly increasing cost distribution G on $[\psi_0, \infty)$, with $\psi_0 > 0$. The most productive firms sit near $\omega = 0$.

A **regulation** changes the marginal-cost profile $\psi \to \psi^R$.

Regulation

Index firms by marginal cost: $\psi_\omega = G^{-1}(\omega)$, for a continuous, strictly increasing cost distribution G on $[\psi_0, \infty)$, with $\psi_0 > 0$. The most productive firms sit near $\omega = 0$.

A **regulation** changes the marginal-cost profile $\psi \to \psi^R$.

Theorem 1 (Which firms gain)

Firm ω's revenue, profit, and market share strictly increase if and only if its relative marginal cost falls:

$$\frac{\psi_\omega^R}{P(\mathbf{p}^R)} < \frac{\psi_\omega}{P(\mathbf{p}^*)}.$$

All three strictly decrease if the inequality is reversed and are unchanged under equality. Where $\mathbf{p}^*$ and $\mathbf{p}^R$ are the equilibrium price profiles.

Proof. Direct application of the envelope theorem $\frac{\partial \pi_\omega^*}{\partial(\psi_\omega/P)} < 0$, and Topkis $\frac{\partial z_\omega^*}{\partial(\psi_\omega/P)} > 0$. → Proof.

Regressive incidence

Proposition 1 (Regressive incidence)

Consider a cost-increasing regulation with incidence $1+\varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega \geq 1$. Suppose incidence is **regressive**:

$$\varepsilon_\omega < \varepsilon_{\omega'} \qquad \text{whenever } \omega < \omega'.$$

Then there exists a cutoff $\bar{\omega} \in (0,1)$ such that every firm in $(0,\bar{\omega})$ strictly gains revenue, profit, and market share, while every firm in $(\bar{\omega},1)$ strictly loses all three.

→ Proofs.

Regressive incidence

Proposition 1 (Regressive incidence)

Consider a cost-increasing regulation with incidence $1+\varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega \geq 1$. Suppose incidence is **regressive**:

$$\varepsilon_\omega < \varepsilon_{\omega'} \qquad \text{whenever } \omega < \omega'.$$

Then there exists a cutoff $\bar{\omega} \in (0,1)$ such that every firm in $(0,\bar{\omega})$ strictly gains revenue, profit, and market share, while every firm in $(\bar{\omega},1)$ strictly loses all three.

Corollary 1 (Exit amplifies)

Suppose firms must pay a fixed operating cost $F > 0$ to remain active. A regressive regulation causes the least productive firms to exit. Thus every surviving firm in $(0,\bar{\omega})$ gains strictly more than under the same regulation without exit.

→ Proofs.

Outline

- A firm gains when a regulation raises the price index proportionally more than its own marginal cost.
- Derive the estimating equation.
- Apply it to FSMA.
- The mechanism strengthens in general equilibrium.
- FSMA's welfare cost is at least $362M a year.

Preview: Elasticity of incidence

Write marginal cost as $\psi_\omega = e/\varphi_\omega$, where e is the price of the common input bundle (fixed) and φ_ω is **baseline productivity**. The object we estimate is the **elasticity of incidence**, how a firm's incidence changes with φ_ω:

$$\tau \equiv -\frac{d\ln(1+\varepsilon_\omega)}{d\ln\varphi_\omega}$$

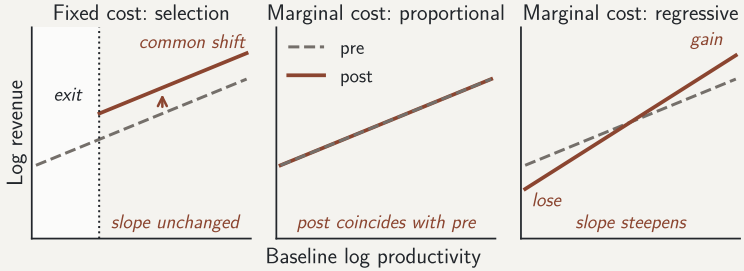
A firm 1% more productive sees its marginal cost increase by $\tau\%$ less.

Preview: Elasticity of incidence

Write marginal cost as $\psi_\omega = e/\varphi_\omega$, where e is the price of the common input bundle (fixed) and φ_ω is **baseline productivity**. The object we estimate is the **elasticity of incidence**, how a firm's incidence changes with φ_ω:

$$\tau \equiv -\frac{d\ln(1+\varepsilon_\omega)}{d\ln\varphi_\omega}$$

A firm 1% more productive sees its marginal cost increase by $\tau\%$ less.



Revenue pre- and post-regulation

In equilibrium firm ω's log revenue is

$$\ln r_\omega^{\text{pre}} = \underbrace{\ln Y + (\sigma - 1) \ln \frac{P}{\mu\, e}}_{\text{common to all firms}} + \underbrace{(\sigma - 1) \ln \varphi_\omega}_{\text{firm specific}},$$

where $\mu = \sigma/(\sigma - 1)$ is the markup.

Revenue pre- and post-regulation

In equilibrium firm ω's log revenue is

$$\ln r_\omega^{\text{pre}} \;=\; A^{\text{pre}} \;+\; \underbrace{(\sigma-1)\ln \varphi_\omega}_{\text{firm specific}}\,,$$

where $\mu = \sigma/(\sigma-1)$ is the markup.

Revenue pre- and post-regulation

In equilibrium firm ω's log revenue is

$$\ln r_\omega^{\text{pre}} \;=\; A^{\text{pre}} \;+\; \underbrace{(\sigma-1)\ln\varphi_\omega}_{\text{firm specific}}\,,$$

where $\mu = \sigma/(\sigma-1)$ is the markup.

A regulation multiplies ω's marginal cost by $(1+\varepsilon_\omega)$, so $\psi_\omega^R = (1+\varepsilon_\omega)\,e/\varphi_\omega$ and

$$\ln r_\omega^{\text{post}} \;=\; \ln Y + (\sigma-1)\ln\frac{P^R}{\mu\,e} \;+\; (\sigma-1)\ln\frac{\varphi_\omega}{1+\varepsilon_\omega}\,.$$

Revenue pre- and post-regulation

In equilibrium firm ω's log revenue is

$$\ln r_\omega^{\text{pre}} \;=\; A^{\text{pre}} \;+\; \underbrace{(\sigma-1)\ln \varphi_\omega}_{\text{firm specific}}\,,$$

where $\mu = \sigma/(\sigma-1)$ is the markup.

A regulation multiplies ω's marginal cost by $(1+\varepsilon_\omega)$, so $\psi_\omega^R = (1+\varepsilon_\omega)\, e/\varphi_\omega$ and

$$\ln r_\omega^{\text{post}} \;=\; A^{\text{post}} \;+\; \underbrace{(\sigma-1)(1+\tau)\ln \varphi_\omega}_{\text{firm specific}}\,, \quad \text{under constant } \tau$$

How steeply revenue changes with productivity

$$\ln r_{\omega}^{\text{post}} \;=\; A^{\text{post}} \;+\; (\sigma - 1)\mathbf{(1 + \tau)} \ln \varphi_{\omega}$$

How steeply revenue changes with productivity

$$\ln r_\omega^{\text{post}} \;=\; A^{\text{post}} \;+\; (\sigma - 1)\mathbf{(1+\tau)} \ln \varphi_\omega$$

A regulation changes how baseline productivity φ_ω relates to revenue:

- If the regulation raises fixed costs ($\varepsilon_\omega = 0$ for all ω), or incidence is proportional ($\varepsilon_\omega = \varepsilon$ for all ω), then $\tau = 0$ and the **slope** is unchanged.

How steeply revenue changes with productivity

$$\ln r_\omega^{\text{post}} \;=\; A^{\text{post}} \;+\; (\sigma - 1)\mathbf{(1+\tau)} \ln \varphi_\omega$$

A regulation changes how baseline productivity φ_ω relates to revenue:

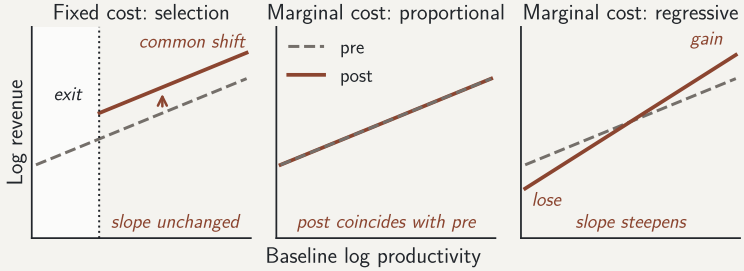
- If the regulation raises fixed costs ($\varepsilon_\omega = 0$ for all ω), or incidence is proportional ($\varepsilon_\omega = \varepsilon$ for all ω), then $\tau = 0$ and the **slope** is unchanged.
- If incidence is regressive, then $\tau > 0$ and the slope **steepens**.

How steeply revenue changes with productivity

$$\ln r_\omega^{\text{post}} \;=\; A^{\text{post}} \;+\; (\sigma - 1)\mathbf{(1+\tau)} \ln \varphi_\omega$$

A regulation changes how baseline productivity φ_ω relates to revenue:

- If the regulation raises fixed costs ($\varepsilon_\omega = 0$ for all ω), or incidence is proportional ($\varepsilon_\omega = \varepsilon$ for all ω), then $\tau = 0$ and the **slope** is unchanged.
- If incidence is regressive, then $\tau > 0$ and the slope **steepens**.



Baseline revenue as a proxy for productivity

Proxy baseline productivity with *baseline revenue* r_ω^0:

$$\ln r_\omega^0 \;=\; A^0 + (\sigma - 1)\ln \varphi_\omega \qquad \Longleftrightarrow \qquad \ln \varphi_\omega \;=\; \frac{\ln r_\omega^0 - A^0}{\sigma - 1}$$

Baseline revenue as a proxy for productivity

Proxy baseline productivity with *baseline revenue* r_ω^0:

$$\ln r_\omega^0 \;=\; A^0 + (\sigma - 1)\ln \varphi_\omega \qquad \Longleftrightarrow \qquad \ln \varphi_\omega \;=\; \frac{\ln r_\omega^0 - A^0}{\sigma - 1}$$

Define $\Delta \ln r_\omega \equiv \ln r_\omega^{\text{post}} - \ln r_\omega^{\text{pre}}$. Then

$$\begin{aligned} \Delta \ln r_\omega \;&=\; \left(A^{\text{post}} - A^{\text{pre}}\right) + (\sigma - 1)\,\tau \ln \varphi_\omega \\ &=\; A^\Delta + \tau \ln r_\omega^0 \end{aligned}$$

With post_t indicating the post-regulation period,

$$\begin{aligned} \ln r_{\omega t} \;&=\; (1 - \text{post}_t)\,\ln r_\omega^{\text{pre}} \;+\; \text{post}_t\,\ln r_\omega^{\text{post}} \\ &=\; \underbrace{\left(A^{\text{pre}} - A^0\right) + A^\Delta \cdot \text{post}_t}_{\delta_t} \;+\; \underbrace{\ln r_\omega^0}_{\delta_\omega} + \tau\left(\ln r_\omega^0 \times \text{post}_t\right) \end{aligned}$$

Estimating equation for the elasticity of incidence τ

Writing β for the coefficient on the interaction:

$$\ln r_{\omega t} \;=\; \beta\left(\ln r_{\omega}^{0} \times \text{post}_{t}\right) \;+\; \delta_{\omega} \;+\; \delta_{t}\,, \qquad \beta = \tau$$

$$\iff \qquad r_{\omega t} \;=\; \exp\Big\{\beta\left(\ln r_{\omega}^{0} \times \text{post}_{t}\right) \;+\; \delta_{\omega} \;+\; \delta_{t}\Big\}$$

Estimating equation for the elasticity of incidence τ

Writing β for the coefficient on the interaction:

$$\ln r_{\omega t} \;=\; \beta\left(\ln r_{\omega}^{0} \times \text{post}_{t}\right) \;+\; \delta_{\omega} \;+\; \delta_{t}\,, \qquad \beta = \tau$$

$$\iff \qquad r_{\omega t} \;=\; \exp\Big\{\beta\left(\ln r_{\omega}^{0} \times \text{post}_{t}\right) \;+\; \delta_{\omega} \;+\; \delta_{t}\Big\}$$

Estimated by Poisson pseudo-maximum likelihood (PPML) (Santos Silva and Tenreyro, 2006):

$$r_{\omega t} \;=\; \exp\Big\{\beta\left(\ln r_{\omega}^{0} \times \text{post}_{t}\right) \;+\; \delta_{\omega} \;+\; \delta_{t}\Big\} \cdot \eta_{\omega t}, \qquad \mathbb{E}\big[\eta_{\omega t} \,|\, \cdot\,\big] = 1$$

General-equilibrium changes in expenditure Y, input prices e, and the price index P are absorbed by the time fixed effects.

Outline

- A firm gains when a regulation raises the price index proportionally more than its own marginal cost.
- Derive the estimating equation.
- Apply it to FSMA with Nielsen scanner data.
- The mechanism strengthens in general equilibrium.
- FSMA's welfare cost is at least $362M a year.

Food Safety Modernization Act

- FSMA, the largest reform of U.S. food-safety law since 1938, took effect in September 2016.
- Its justification came from the 2008–09 *Salmonella* outbreak at the Peanut Corporation of America (PCA).

Food Safety Modernization Act

- FSMA, the largest reform of U.S. food-safety law since 1938, took effect in September 2016.
- Its justification came from the 2008–09 *Salmonella* outbreak at the Peanut Corporation of America (PCA).
- Yet the resulting regulation would not have prevented the outbreak.
- Instead, the act largely **codified the practices of the firms lobbying for it**.

Food Safety Modernization Act

- FSMA, the largest reform of U.S. food-safety law since 1938, took effect in September 2016.
- Its justification came from the 2008–09 *Salmonella* outbreak at the Peanut Corporation of America (PCA).
- Yet the resulting regulation would not have prevented the outbreak.
- Instead, the act largely **codified the practices of the firms lobbying for it**.
- The largest manufacturers already ran the controls FSMA made mandatory: monitoring, sanitizing, and record-keeping.
- For smaller firms, it meant new personnel and new programs that recur with every production run: **a marginal-cost increase**.

Food categories

- **Sample:** dry, ready-to-eat foods exposed to *Salmonella*.
- Long treated as low-risk, these foods came under mandatory controls only with FSMA.

Food categories

- **Sample:** dry, ready-to-eat foods exposed to *Salmonella*.
- Long treated as low-risk, these foods came under mandatory controls only with FSMA.
- **Design:** compare products **within** a category. They share the shelf and face similar shocks, but only some carry the FSMA obligation.

Food categories

- **Sample:** dry, ready-to-eat foods exposed to *Salmonella*.
- Long treated as low-risk, these foods came under mandatory controls only with FSMA.
- **Design:** compare products **within** a category. They share the shelf and face similar shocks, but only some carry the FSMA obligation.
- Example, nuts:

Treated: roasted nuts

The firm must control contamination in its own facility.

Placebo: raw nuts

Control is shifted to the farmer.

Data

- 2012–2019 Nielsen scanner data: sales by store $\times$ week $\times$ barcode.
- Each barcode is linked to its parent firm, then summed over the weeks in a quarter and over the stores in a county.
- Unit of observation: revenue by firm $\times$ county $\times$ quarter, within a product (e.g., raw nuts).

→ More on the data.

Data

- 2012–2019 Nielsen scanner data: sales by store $\times$ week $\times$ barcode.
- Each barcode is linked to its parent firm, then summed over the weeks in a quarter and over the stores in a county.
- Unit of observation: revenue by firm $\times$ county $\times$ quarter, within a product (e.g., raw nuts).
- Observations where the firm exits the market are dropped.
- Zeros that capture retreat from a county are kept.

→ More on the data.

Estimation

The elasticity of incidence is estimated by PPML:

$$r_{ict} \;=\; \exp\Big\{\beta\left(\ln r_i^0 \times \text{post}_t\right) \;+\; \delta_{ic} \;+\; \delta_{st}\Big\} \cdot \eta_{ict}, \qquad \mathbb{E}\big[\eta_{ict} \,|\, \cdot\,\big] = 1$$

where i indexes firms, c counties, s states, and t quarters; δ_{ic} are firm $\times$ county and δ_{st} state $\times$ quarter fixed effects.

- **Baseline size** r_i^0: the firm's 2011 revenue in the product (the year before the estimation window).

Estimation

The elasticity of incidence is estimated by PPML:

$$r_{ict} \;=\; \exp\Big\{\beta\left(\ln r_i^0 \times \text{post}_t\right) \;+\; \delta_{ic} \;+\; \delta_{st}\Big\} \cdot \eta_{ict}, \qquad \mathbb{E}\big[\eta_{ict} \,|\, \cdot\,\big] = 1$$

where i indexes firms, c counties, s states, and t quarters; δ_{ic} are firm $\times$ county and δ_{st} state $\times$ quarter fixed effects.

- **Baseline size** r_i^0: the firm's 2011 revenue in the product (the year before the estimation window).
- Compliance began in September 2016, so 2016 is dropped (a donut design). Pre-period 2012–2015; post-period 2017–2019.
- Where the event study shows a linear pre-trend, it is removed by extrapolation, and the trend break and the trend-adjusted post average estimates are reported instead (Dobkin et al., 2018; Freyaldenhoven et al., 2026).

Results: products with no pre-trends

Product	Role	$\hat{\beta}$	s.e.	Obs. (000s)	Firms
Nuts, oil-roasted	Treated	$+0.456^{***}$	(0.092)	518	24
Nuts, dry-roasted	Treated	$+0.238^{***}$	(0.077)	604	34
Nuts, raw	Placebo	-0.075	(0.106)	380	36
Dried fruit, true-dried	Treated	$+0.292^{***}$	(0.096)	424	20
Dried fruit, fruit snacks	Placebo	-0.059	(0.065)	516	18
Peanut butter, creamy	Treated	$+0.103^{***}$	(0.039)	421	14
Peanut butter, chunky	Treated	$+0.097^{**}$	(0.050)	284	12

Donut difference-in-differences by PPML, 2012–2015 vs. 2017–2019. Wild cluster bootstrap by firm. $^{**}p < 0.05$, $^{***}p < 0.01$.

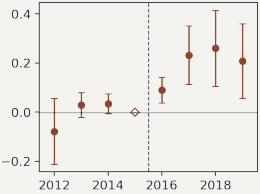
Oil-roasted nuts: two firms with a 10:1 revenue ratio in 2011 ended the post period near 30:1 (post ratio: $10 \times 10^{0.456} = 28.6$).

Event studies: nuts and dried fruit

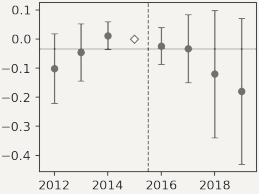
Nuts, oil-roasted



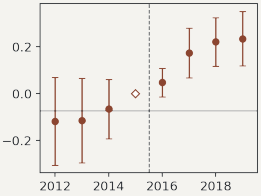
Nuts, dry-roasted



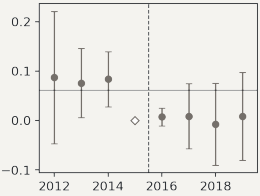
Nuts, raw (placebo)



Dried fruit, true-dried



Dried fruit, fruit snacks (placebo)



Estimates of the elasticity of incidence by calendar year, with 95% confidence intervals. Compliance began in September 2016.

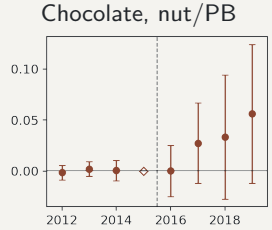
→ Remaining event studies.

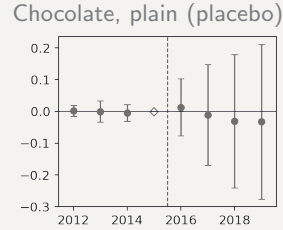
Results: products with linear pre-trends

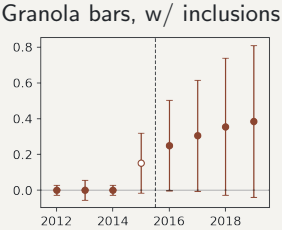
Product	Role	Trend break (/yr)	s.e.	Post avg.	Obs. (000s)	Firms
Chocolate, nut/PB	Treated	$+0.014^{*}$	(0.008)	$+0.039$	1,041	35
Chocolate, plain	Placebo	-0.011	(0.026)	-0.025	681	27
Granola bars, w/ inclusions	Treated	$+0.035$	(0.041)	$+0.348^{*}$	382	15
Granola bars, plain	Placebo	-0.046	(0.041)	-0.073	296	10
Granola cereal	Treated	$+0.112^{***}$	(0.032)	$+0.256^{**}$	317	12
Cereal	Treated	$+0.036^{***}$	(0.008)	$+0.064^{***}$	549	25
Seasoning	Treated	$+0.015^{***}$	(0.006)	$+0.049^{***}$	840	35
Crackers, flavored	Treated	$+0.024^{**}$	(0.012)	$+0.027$	621	23

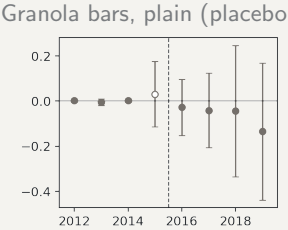
Linear pre-trend fitted on 2012–2015 and removed by extrapolation. Trend break: change per year at the event. Post avg.: 2017–2019 level net of the extrapolated pre-trend. Wild cluster bootstrap by firm. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Event studies: chocolate and granola bars









Elasticity of incidence by calendar year, net of the linear pre-trend, with 95% confidence intervals. Compliance began in September 2016.

→ Remaining event studies.

Outline

- A firm gains when a regulation raises the price index proportionally more than its own marginal cost.
- Derived the estimating equation for the elasticity of incidence and applied it to FSMA: large firms grew relative to small firms in affected products, but not in placebos.
- Bring the mechanism to general equilibrium.
  It strengthens when demand across industries is inelastic.
- Quantify FSMA's welfare cost.
  The cost is at least $362M a year.

General Equilibrium

- The mechanism is embedded in a nested CES economy with endogenous expenditure and input price.

General Equilibrium

- The mechanism is embedded in a nested CES economy with endogenous expenditure and input price.
- Under the standard inelastic outer nest assumption ($\sigma_0 \leq 1$), the industry-wide price increase **draws in expenditure from other industries**.
- So the top firms of the regulated industry **gain even more** revenue and profit than in partial equilibrium.

General Equilibrium

- The mechanism is embedded in a nested CES economy with endogenous expenditure and input price.
- Under the standard inelastic outer nest assumption ($\sigma_0 \leq 1$), the industry-wide price increase **draws in expenditure from other industries**.
- So the top firms of the regulated industry **gain even more** revenue and profit than in partial equilibrium.
- The same model and the elasticity of incidence estimates measure a lower bound on the welfare cost.

Environment

- A representative household supplies L units of labor inelastically and owns all firms.
- Preferences over industries $h = 1, \ldots, H$ are an outer CES nest with elasticity σ_0 and weights α_h; within each industry h, preferences are CES with elasticity σ_h.

$$U = \left(\sum_h \alpha_h^{\frac{1}{\sigma_0}} C_h^{\frac{\sigma_0 - 1}{\sigma_0}} \right)^{\frac{\sigma_0}{\sigma_0 - 1}}, \qquad C_h = \left(\int x_\omega^{\frac{\sigma_h - 1}{\sigma_h}} \, d\omega \right)^{\frac{\sigma_h}{\sigma_h - 1}}$$

- Labor is the only factor, and the wage $w = 1$ is the numeraire.
- The household spends its income Y, which consists of wages and profits.

Environment

Expenditure shares S_h and the aggregate price index P_0 are:

$$S_h = \frac{\alpha_h P_h^{1-\sigma_0}}{\sum_k \alpha_k P_k^{1-\sigma_0}}, \qquad P_0 = \left(\sum\nolimits_h \alpha_h P_h^{1-\sigma_0}\right)^{\frac{1}{1-\sigma_0}},$$

where P_h is industry h's price aggregator.

- A firm in industry h charging price p earns revenue $r = (p/P_h)^{1-\sigma_h} S_h Y$.
- As in partial equilibrium, the regulated industry g has inner elasticity $\sigma_g > 1$, with a unit mass of firms.
- A firm with productivity φ uses $1/\varphi$ units of labor per unit of output, so its marginal cost is $\psi = w/\varphi \in [\psi_{\min}, \infty)$.

The Regulation

- The regulation takes the form of unproductive labor. A firm producing q units hires $\ell_p = q/\varphi$ units of productive labor and $\ell_c = \varepsilon(\psi)\, q/\varphi$ units of unproductive labor (e.g., the labor needed to monitor, sanitize, and keep records, as in FSMA).

The Regulation

- The regulation takes the form of unproductive labor. A firm producing q units hires $\ell_p = q/\varphi$ units of productive labor and $\ell_c = \varepsilon(\psi)\, q/\varphi$ units of unproductive labor (e.g., the labor needed to monitor, sanitize, and keep records, as in FSMA).
- The labor requirement increases to $(1 + \varepsilon(\psi))/\varphi$ per unit of output, so marginal cost becomes:

$$\psi^R \;=\; \big(1 + \varepsilon(\psi)\big)\, \psi$$

- Incidence is regressive: $\varepsilon(\psi)$ is continuous and strictly increasing in ψ, with $\varepsilon(\psi_{\min}) \geq 0$.

Income

- Under CES, industry h's profit margin is $1/\sigma_h$.
- Write $m_g \equiv 1/\sigma_g$ for the regulated industry's margin, and define the expenditure-weighted average margin outside g:

$$m_{-g} \equiv \sum_{h \neq g} \frac{S_h}{1 - S_g} \frac{1}{\sigma_h}$$

- The regulation leaves m_{-g} unchanged, since all shares outside g scale by the same factor.

Income

- Under CES, industry h's profit margin is $1/\sigma_h$.
- Write $m_g \equiv 1/\sigma_g$ for the regulated industry's margin, and define the expenditure-weighted average margin outside g:

$$m_{-g} \equiv \sum_{h \neq g} \frac{S_h}{1 - S_g} \frac{1}{\sigma_h}$$

- The regulation leaves m_{-g} unchanged, since all shares outside g scale by the same factor.
- Income Y depends on the profit composition of spending:

$$Y = \frac{wL}{1 - \kappa}, \qquad \kappa \equiv S_g \, m_g + (1 - S_g) \, m_{-g}$$

- It increases ($Y^R > Y$) if $m_g > m_{-g}$, as profits shift toward the higher-margin industry. It decreases otherwise.

The regulation draws in expenditure

The result below holds regardless of what happens to Y, given $\sigma_0 \leq 1$.

The regulation draws in expenditure

The result below holds regardless of what happens to Y, given $\sigma_0 \leq 1$.

Lemma 1 (Industry expenditure never decreases)

After the regulation, the log change in expenditure on industry g satisfies

$$B \equiv \Delta \ln(S_g Y) \geq 0,$$

with equality only when $\sigma_0 = 1$. → Proof.

So the regulation draws expenditure into the regulated industry. Even if Y decreases, the increase in S_g more than offsets it.

→ $\sigma_0 > 1$.

Revenue within the regulated industry

Revenue pre- and post-regulation is:

$$\begin{aligned}\ln r(\psi) &= (1-\sigma_g)\big[\ln \mu_g + \ln \psi - \ln P_g\big] + \ln S_g + \ln Y \\ \ln r^R(\psi) &= (1-\sigma_g)\big[\ln \mu_g + \ln \psi + \ln(1+\varepsilon(\psi)) - \ln P_g^R\big] + \ln S_g^R + \ln Y^R\end{aligned}$$

So:

$$\Delta \ln r(\psi) = (\sigma_g - 1)\big[\Delta \ln P_g - \ln(1+\varepsilon(\psi))\big] + B$$

$\Delta \ln r(\psi)$ is strictly decreasing in ψ.

Which firms gain in general equilibrium

Theorem 2 (Which firms gain in general equilibrium)

A firm with cost ψ gains revenue and profit if and only if

$$\frac{\psi^R}{P_g^R} < \frac{\psi}{P_g} e^{B/(\sigma_g - 1)}.$$

The most productive firms always gain revenue and profit. → Proof.

→ $\sigma_0 > 1$.

Which firms gain in general equilibrium

Theorem 2 (Which firms gain in general equilibrium)

A firm with cost ψ gains revenue and profit if and only if

$$\frac{\psi^R}{P_g^R} < \frac{\psi}{P_g} e^{B/(\sigma_g - 1)}.$$

The most productive firms always gain revenue and profit. → Proof.

Recall that the partial equilibrium condition for gaining is $\frac{\psi^R}{P_g^R} < \frac{\psi}{P_g}$. In general equilibrium, the regulation brings in expenditure B, so the set of growing firms expands, and firms that gain in partial equilibrium gain even more.

→ $\sigma_0 > 1$.

Outline

- A firm gains when a regulation raises the price index proportionally more than its own marginal cost.
- Derived the estimating equation for the elasticity of incidence and applied it to FSMA: large firms grew relative to small firms in affected products, but not in placebos.
- The mechanism strengthens in general equilibrium.
- Quantify FSMA's welfare cost.
  The cost is at least $362M a year.

- What would a researcher using the homogeneous-goods tradition conclude about welfare?

Pareto marginal costs

- Assume marginal costs are Pareto, with shape θ and scale $\psi_{\min}$:

$$\Pr(\psi_\omega > \psi) \;=\; \left(\frac{\psi_{\min}}{\psi}\right)^{\theta}, \qquad \psi \in [\psi_{\min}, \infty).$$

Pareto marginal costs

- Assume marginal costs are Pareto, with shape θ and scale $\psi_{\min}$:

$$\Pr(\psi_\omega > \psi) \;=\; \left(\frac{\psi_{\min}}{\psi}\right)^{\theta}, \qquad \psi \in [\psi_{\min}, \infty).$$

- With a constant elasticity of incidence, incidence can be written as

$$1 + \varepsilon(\psi) \;=\; \underbrace{\left(1 + \varepsilon(\psi_{\min})\right)}_{\text{proportional}} \underbrace{\left(\frac{\psi}{\psi_{\min}}\right)^{\tau}}_{\text{regressive component}}.$$

Pareto marginal costs

- Assume marginal costs are Pareto, with shape θ and scale $\psi_{\min}$:

$$\Pr(\psi_\omega > \psi) \;=\; \left(\frac{\psi_{\min}}{\psi}\right)^{\theta}, \qquad \psi \in [\psi_{\min}, \infty).$$

- With a constant elasticity of incidence, incidence can be written as

$$1 + \varepsilon(\psi) \;=\; \underbrace{\left(1 + \varepsilon(\psi_{\min})\right)}_{\text{proportional}} \underbrace{\left(\frac{\psi}{\psi_{\min}}\right)^{\tau}}_{\text{regressive component}}.$$

- The elasticity of incidence τ changes the shape θ, and the proportional component $1 + \varepsilon(\psi_{\min})$ changes the scale $\psi_{\min}$:

$$\Pr\left(\psi^R_\omega > \psi\right) \;=\; \left(\frac{\left(1 + \varepsilon(\psi_{\min})\right)\psi_{\min}}{\psi}\right)^{\frac{\theta}{1+\tau}}, \qquad \psi \in \left[\left(1 + \varepsilon(\psi_{\min})\right)\psi_{\min}, \infty\right).$$

→

Welfare

- The proportional component of incidence is not estimated, so set $\varepsilon(\psi_{\min}) = 0$. It would have increased the welfare cost, the expenditure B drawn into the industry, and $\Delta \ln P_g$. So the estimates below are lower bounds.

Welfare

- The proportional component of incidence is not estimated, so set $\varepsilon(\psi_{\min}) = 0$. It would have increased the welfare cost, the expenditure B drawn into the industry, and $\Delta \ln P_g$. So the estimates below are lower bounds.
- Welfare is real income, $W = Y/P_0$. Assume a common margin ($m_g = m_{-g}$):

$$\Delta \ln W \;=\; -\Delta \ln P_0 \;=\; -\frac{1}{1-\sigma_0}\,\ln\Big[1 - S_g + S_g\, e^{(1-\sigma_0)\,\Delta \ln P_g}\Big] \;<\; 0.$$

$\rightarrow m_g \neq m_{-g}$.

Welfare

- The proportional component of incidence is not estimated, so set $\varepsilon(\psi_{\min}) = 0$. It would have increased the welfare cost, the expenditure B drawn into the industry, and $\Delta \ln P_g$. So the estimates below are lower bounds.
- Welfare is real income, $W = Y/P_0$. Assume a common margin ($m_g = m_{-g}$):

$$\Delta \ln W \;=\; -\Delta \ln P_0 \;=\; -\frac{1}{1-\sigma_0} \ln\Big[1 - S_g + S_g\, e^{(1-\sigma_0)\,\Delta \ln P_g}\Big] \;<\; 0.$$

$\rightarrow m_g \neq m_{-g}$.

- Pareto gives a closed form for the change in the industry price index:

$$\Delta \ln P_g \;=\; \frac{1}{\sigma_g - 1} \ln \frac{\theta + (1+\tau)(\sigma_g - 1)}{\theta + \sigma_g - 1}.$$

Welfare

- The proportional component of incidence is not estimated, so set $\varepsilon(\psi_{\min}) = 0$. It would have increased the welfare cost, the expenditure B drawn into the industry, and $\Delta \ln P_g$. So the estimates below are lower bounds.
- Welfare is real income, $W = Y/P_0$. Assume a common margin ($m_g = m_{-g}$):

$$\Delta \ln W \;=\; -\Delta \ln P_0 \;=\; -\frac{1}{1-\sigma_0}\,\ln\Big[\,1 - S_g + S_g\, e^{(1-\sigma_0)\,\Delta \ln P_g}\Big] \;<\; 0.$$

$\rightarrow m_g \neq m_{-g}$.

- Pareto gives a closed form for the change in the industry price index:

$$\Delta \ln P_g \;=\; \frac{1}{\sigma_g - 1}\,\ln \frac{\theta + (1+\tau)(\sigma_g - 1)}{\theta + \sigma_g - 1}.$$

- Given $(\theta, \sigma_g, \sigma_0, S_g)$, the welfare cost estimate from the regressive component is a closed-form function of $\hat{\beta}$.

Parameters

- $\hat{\beta}$: the product-specific FSMA estimates of the elasticity of incidence.
- θ: the pre-regulation Pareto shape, estimated from 2011 baseline revenues with the Hill tail-index estimator (Hill, 1975).
- $\sigma_0 = 0.5$: based on category-level food price elasticities of 0.27–0.81 across the studies reviewed by Andreyeva et al. (2010).
- $\sigma_g = 3.9$: the median elasticity of substitution within an industry, estimated by Hottman et al. (2016) from Nielsen data.
- S_g: the product's Nielsen revenue, scaled to the national market and divided by personal consumption expenditure.

→ Sensitivity to σ_0 and σ_g.

Quantification by treated product

Product	$\hat{\beta}$	θ	$\Delta \ln P_g$ (%)	B (%)	Firms gaining (%)	Welfare cost ($M/yr)
Nuts, oil-roasted	0.456	0.89	10.3	5.2	21	41
Nuts, dry-roasted	0.238	0.86	5.8	2.9	22	44
Dried fruit, true-dried	0.292	0.80	7.1	3.6	20	19
Peanut butter, creamy	0.103	0.74	2.7	1.4	21	26
Peanut butter, chunky	0.097	0.79	2.5	1.3	21	5
Chocolate, nut/PB	0.039	0.57	1.1	0.6	17	27
Granola bars, w/ inclusions	0.348	0.74	8.4	4.2	19	56
Granola cereal	0.256	1.26	5.7	2.8	28	12
Cereal	0.064	0.54	1.8	0.9	16	105
Seasoning	0.049	0.64	1.4	0.7	19	15
Crackers, flavored	0.027	0.57	0.8	0.4	17	11

In oil-roasted nuts, the product's price index rises by 10%, expenditure on it by 5%, the winners are the most productive fifth of firms, and the annual welfare cost is $41 million.

In every product, between a sixth and a third of firms gain. Across the eleven treated products, annual welfare costs total $362 million.

→ Standard errors and confidence intervals.

The homogeneous-goods benchmark

- What would a researcher using the *raising rivals' costs* tradition conclude about welfare?
- Take the simplest model in that tradition: a homogeneous good supplied by a low-cost leader and n identical higher-cost followers.
- They compete à la Cournot under linear demand $p = a - bQ$.
- The regulation raises the followers' marginal cost c_f and spares the leader's, c_L.
- Welfare is total surplus: consumer plus producer surplus.

The homogeneous-goods benchmark

- What would a researcher using the *raising rivals' costs* tradition conclude about welfare?
- Take the simplest model in that tradition: a homogeneous good supplied by a low-cost leader and n identical higher-cost followers.
- They compete à la Cournot under linear demand $p = a - bQ$.
- The regulation raises the followers' marginal cost c_f and spares the leader's, c_L.
- Welfare is total surplus: consumer plus producer surplus.

In four of the studied products, this model would report welfare gains instead of losses.

→ Conclusion.

Why does the welfare result reverse?

- Given demand $p = a - bQ$ and the number of followers n, the leader's share

$$s_L = \frac{(a - c_L) + n\,(c_f - c_L)}{(n+1)(a - c_L) - n\,(c_f - c_L)}$$

  is an increasing function of the cost gap $c_f - c_L$. This lets us determine the sign of the welfare change knowing only the leader's share and the number of firms.

Why does the welfare result reverse?

- Given demand $p = a - bQ$ and the number of followers n, the leader's share

$$s_L = \frac{(a - c_L) + n\,(c_f - c_L)}{(n+1)(a - c_L) - n\,(c_f - c_L)}$$

  is an increasing function of the cost gap $c_f - c_L$. This lets us determine the sign of the welfare change knowing only the leader's share and the number of firms.

- An increase in the followers' cost changes total surplus in two opposite ways:
  (1) the followers pay a higher cost on every unit they produce;
  (2) output reallocates from the higher-cost followers to the low-cost leader, **a gain in efficiency**.

The threshold

Proposition 2 (When the homogeneous model reports a gain)

An increase in the followers' cost c_f raises total surplus if

$$s_L > s_L^*(n) \equiv \frac{n+4}{2(n+2)} \in \left(\tfrac{1}{2}, \tfrac{5}{6}\right].$$

→ Proof.

The threshold

Proposition 2 (When the homogeneous model reports a gain)

An increase in the followers' cost c_f raises total surplus if

$$s_L > s_L^*(n) \equiv \frac{n+4}{2(n+2)} \in \left(\tfrac{1}{2}, \tfrac{5}{6}\right].$$

At the firm counts of the FSMA application, the leader's share is above its threshold in:

Product	Leader	Share	Threshold
Nuts, dry-roasted	Wonderful	0.64	0.53
Crackers, flavored	Mondelēz	0.63	0.54
Seasoning	McCormick	0.56	0.53
Peanut butter, creamy	Smucker's (Jif)	0.57	0.57

→ Proof.

Conclusion

- Firms can benefit from a regulation that raises production costs, including their own.
- When incidence is non-proportional, every price rises, but the least burdened firms raise theirs the least.
- They become relatively cheaper, attract demand, and gain revenue, profit, and market share.

Conclusion

- Firms can benefit from a regulation that raises production costs, including their own.
- When incidence is non-proportional, every price rises, but the least burdened firms raise theirs the least.
- They become relatively cheaper, attract demand, and gain revenue, profit, and market share.
- The mechanism does not require exemptions, barriers to entry, or the forced exit of rivals.

Conclusion

- Evidence of this mechanism in FSMA, using Nielsen scanner data.
- Large firms grow relative to small firms in products where FSMA binds, but not in similar placebo products.
- The reallocation is economically large: in oil-roasted nuts, a 10:1 revenue ratio between two firms before the regulation became nearly 30:1 after it.
- The largest firms in these categories supported and lobbied for the act.

Conclusion

- With inelastic substitution across industries, general equilibrium strengthens the mechanism: expenditure is drawn in from other industries.
- Calibrated with the FSMA estimates, the model implies that only the top sixth to third of firms gain, and that the welfare cost is at least $362 million a year.

Conclusion

- With inelastic substitution across industries, general equilibrium strengthens the mechanism: expenditure is drawn in from other industries.
- Calibrated with the FSMA estimates, the model implies that only the top sixth to third of firms gain, and that the welfare cost is at least $362 million a year.
- These results change how regulation should be evaluated: its incidence schedule is itself an object of political economy.
- This new mechanism proposes that by changing relative prices, regulation reallocates demand. Effectively picking winners and losers.

Thank you

My Website

- Comments and questions welcome
- Working papers available at: gabriel.cr
- Contact: gabrielcr@ucsd.edu

Preliminary Results on Prices: Nuts

- Average price paid per ounce: total revenue over total ounces sold, by product and year.
- Sample: the firms of the estimation (24 oil-roasted, 34 dry-roasted, 36 raw), balanced stores.
- Raw nuts are the placebo: they share the input and the shelf, so the common movement in nut prices drops out of the difference.

Change in average price per ounce (log points)	2012–2015	2015–2019
Oil-roasted	$+18.7$	$+0.5$
Raw	$+21.8$	-10.4
Oil-roasted − raw	$\mathbf{-3.1}$	$\mathbf{+10.9}$
Dry-roasted	$+23.3$	-1.5
Raw	$+21.8$	-10.4
Dry-roasted − raw	$\mathbf{+1.5}$	$\mathbf{+8.9}$

Environment: Consumers (HSA)

Consider a static economy with a unit continuum of differentiated varieties $\omega \in (0, 1)$.

- Consumers have homothetic preferences with a single aggregator (HSA) (Matsuyama and Ushchev, 2017). Expenditure Y is fixed.
- Demand for variety ω:

$$x_\omega \;=\; \frac{Y}{p_\omega}\, s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right)$$

  s_ω is the share function, p_ω the price of variety ω, and $\mathbf{p}$ the price profile. $P(\mathbf{p})$ is the price aggregator, defined implicitly by the adding-up condition:

$$\int_0^1 s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right) d\omega \;=\; 1$$

- Goods are gross substitutes, $s'_\omega(z) < 0$ ($\sigma > 1$ in CES).

Environment: Firms (HSA)

- Each variety ω is produced by a single firm with constant marginal cost $\psi_\omega > 0$, which takes P as given and chooses its price to maximize profit:

$$\pi_\omega \;=\; (p_\omega - \psi_\omega)\, x_\omega \;=\; \left(1 - \frac{\psi_\omega}{p_\omega}\right) Y\, s_\omega\!\left(\frac{p_\omega}{P(\mathbf{p})}\right)$$

- Write $z_\omega \equiv p_\omega / P(\mathbf{p})$ for the relative price. The same problem becomes:

$$\pi_\omega \;=\; \left(1 - \frac{\psi_\omega / P(\mathbf{p})}{z_\omega}\right) Y\, s_\omega(z_\omega)$$

- Maximizing over price levels is the same as maximizing over relative prices. Marginal cost enters the problem only through **relative marginal cost** ψ_ω / P.

Relative marginal cost ψ_ω / P is the sufficient statistic: it determines the firm's relative price, market share, revenue, and profit.

Regulation (HSA)

Index firms by marginal cost: $\psi_\omega = G^{-1}(\omega)$, for a continuous, strictly increasing cost distribution G on $[\psi_0, \infty)$, with $\psi_0 > 0$. The most productive firms sit near $\omega = 0$.

A **regulation** changes the marginal-cost profile $\psi \to \psi^R$.

Theorem 1 (Which firms gain)

Firm ω's revenue, profit, and market share strictly increase if and only if its relative marginal cost falls:

$$\frac{\psi^R_\omega}{P(\boldsymbol{\mu}^R \boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})}.$$

All three strictly decrease if the inequality is reversed and are unchanged under equality. Where $\boldsymbol{p}^* = \boldsymbol{\mu}\boldsymbol{\psi}$ is the equilibrium price profile.

Proof. Direct application of the envelope theorem $\frac{\partial \pi^*_\omega}{\partial(\psi_\omega/P)} < 0$, and Topkis $\frac{\partial z^*_\omega}{\partial(\psi_\omega/P)} > 0$. → Proof.

Regressive incidence (HSA)

Proposition 1 (Regressive incidence)

Consider a cost-increasing regulation with incidence $1+\varepsilon_\omega \equiv \psi^R_\omega/\psi_\omega \geq 1$. Suppose incidence is **regressive**:

$$\varepsilon_\omega < \varepsilon_{\omega'} \qquad \text{whenever } \omega < \omega'.$$

Then there exists a cutoff $\bar{\omega} \in (0,1)$ such that every firm in $(0,\bar{\omega})$ strictly gains revenue, profit, and market share, while every firm in $(\bar{\omega},1)$ strictly loses all three.

Corollary 1 (Exit amplifies)

Suppose firms must pay a fixed operating cost $F > 0$ to remain active. A regressive regulation causes the least productive firms to exit. Thus every surviving firm in $(0,\bar{\omega})$ gains strictly more than under the same regulation without exit.

Proof of Theorem 1 (Which firms gain)

Step 1 (profit). By the envelope theorem, at the equilibrium relative price z^*_ω,

$$\frac{\partial \pi^*_\omega}{\partial(\psi_\omega/P)} = -\frac{Y\, s_\omega(z^*_\omega)}{z^*_\omega} < 0.$$

Step 2 (relative price). Profit has strictly increasing differences in z and ψ_ω/P: the cross-partial is $s_\omega(z)/z^2 - s'_\omega(z)/z > 0$ under gross substitutes. By Topkis, z^*_ω is strictly increasing in ψ_ω/P.

Step 3 (revenue and share). Revenue $Y s_\omega(z^*_\omega)$ and share $s_\omega(z^*_\omega)$ are strictly decreasing in ψ_ω/P, since $s'_\omega < 0$.

Step 4. All three outcomes improve if and only if relative marginal cost falls:

$$\frac{\psi^R_\omega}{P(\boldsymbol{\mu}^R\boldsymbol{\psi}^R)} < \frac{\psi_\omega}{P(\boldsymbol{\mu}\boldsymbol{\psi})} \iff \frac{\psi^R_\omega}{\psi_\omega} < \frac{P(\boldsymbol{\mu}^R\boldsymbol{\psi}^R)}{P(\boldsymbol{\mu}\boldsymbol{\psi})}.$$

←

Proof of Proposition 1 and Corollary 1

Aggregator. By adding-up and gross substitutes, the equilibrium aggregator is strictly increasing in the cost profile and linearly homogeneous: $P(\lambda\boldsymbol{\mu}\boldsymbol{\psi}) = \lambda\, P(\boldsymbol{\mu}\boldsymbol{\psi})$.

Winners. Let $1+\varepsilon_0 \equiv \lim_{\omega\downarrow 0}(1+\varepsilon_\omega)$. Every $\psi_\omega^R \geq (1+\varepsilon_0)\psi_\omega$, strictly on a set of positive measure, so

$$\rho \;\equiv\; \frac{P(\boldsymbol{\mu}^R\boldsymbol{\psi}^R)}{P(\boldsymbol{\mu}\boldsymbol{\psi})} \;>\; 1+\varepsilon_0.$$

Since $\psi_\omega^R/\psi_\omega \to 1+\varepsilon_0 < \rho$ as $\omega \downarrow 0$, Theorem 1 gives gains near the top; by monotonicity the winners are $\{\omega : 1+\varepsilon_\omega < \rho\} = (0, \bar{\omega})$.

Losers. Symmetrically, with $1+\varepsilon_1 \equiv \lim_{\omega\uparrow 1}(1+\varepsilon_\omega)$, $\rho < 1+\varepsilon_1$, so firms near $\omega = 1$ lose, and the losers are $(\bar{\omega}, 1)$.

Corollary 1. Exit shrinks the set of active firms. At the no-exit aggregator, adding-up over the smaller set falls below one; since each share increases in P, the aggregator rises further. Every surviving winner's relative cost falls by more, and F is unchanged. ←

Proof of Lemma 1

Let $\nu \equiv (P_g^R/P_g)^{1-\sigma_0}$ and $D \equiv 1 - S_g + S_g\nu$. The outer nest gives

$$S_g^R = \frac{S_g\,\nu}{D}, \qquad 1 - S_g^R = \frac{1 - S_g}{D}.$$

With $Y = wL/(1-\kappa)$,

$$B \;=\; \Delta \ln S_g + \Delta \ln Y \;=\; \ln\big[\nu(1-\kappa)\big] - \ln\big[D(1-\kappa^R)\big].$$

Substituting S_g^R into κ^R, the sign of B is the sign of

$$\nu(1-\kappa) - D(1-\kappa^R) \;=\; (1-S_g)(\nu-1)(1-m_{-g}).$$

Each factor is nonnegative: $1 - m_{-g} > 0$ because every $\sigma_h > 1$, and $\nu - 1 > 0$ exactly when $\sigma_0 < 1$ (the index rises). So $B \geq 0$, with equality only at $\sigma_0 = 1$. ←

Proof of Theorem 2

Condition. In $\Delta \ln r(\psi) = (\sigma_g - 1)\big[\Delta \ln P_g - \ln(1+\varepsilon(\psi))\big] + B$, the terms $\Delta \ln P_g$ and B are common to all firms. A firm gains iff $\Delta \ln r(\psi) > 0$, that is, iff

$$\ln(1+\varepsilon(\psi)) - \Delta \ln P_g \;<\; \frac{B}{\sigma_g - 1} \quad\Longleftrightarrow\quad \frac{\psi^R}{P_g^R} \;<\; \frac{\psi}{P_g}\, e^{B/(\sigma_g - 1)},$$

using $\psi^R/\psi = 1 + \varepsilon(\psi)$. Profit is r/σ_g, so it moves with revenue.

Monotonicity. $\ln(1+\varepsilon(\psi))$ is strictly increasing in ψ, so $\Delta \ln r(\psi)$ is strictly decreasing.

The most productive firms gain. The index change is a power mean of the incidences:

$$\Big(\frac{P_g^R}{P_g}\Big)^{1-\sigma_g} \;=\; \int s(\psi)\big(1+\varepsilon(\psi)\big)^{1-\sigma_g}\, dG(\psi) \;<\; \big(1+\varepsilon(\psi_{\min})\big)^{1-\sigma_g},$$

so $\Delta \ln P_g > \ln(1+\varepsilon(\psi_{\min}))$. With $B \geq 0$ (Lemma 1), $\Delta \ln r(\psi_{\min}) > 0$.

←

Proof of Theorem 3

The index rises. As in Theorem 2, the power mean gives $\Delta \ln P_g > \ln(1 + \varepsilon(\psi_{\min})) = 0$.

An identity. With $\nu \equiv (P_g^R / P_g)^{1-\sigma_0}$, $D \equiv 1 - S_g + S_g \nu$, and $\Delta \ln S_g = (1 - \sigma_0) \Delta \ln P_g - \ln D$,

$$\Delta \ln r(\psi_{\min}) \;=\; (\sigma_g - \sigma_0)\, \Delta \ln P_g \;-\; \ln D \;+\; \Delta \ln Y.$$

Common margin. $\Delta \ln Y = 0$. Since $\sigma_0 > 1$ and $\Delta \ln P_g > 0$, $\nu < 1$ and $D < 1$, so $-\ln D > 0$ and

$$\Delta \ln r(\psi_{\min}) \;>\; (\sigma_g - \sigma_0)\, \Delta \ln P_g \;>\; 0.$$

Heterogeneous margins. If $\sigma_h \geq 2$ for every h, then $|\Delta \ln Y| \leq -\ln D$, so $-\ln D + \Delta \ln Y \geq 0$ and the same bound holds.

Single crossing. As in Theorem 2, $\Delta \ln r(\psi)$ is strictly decreasing, so the winners are a top segment.

←

Proof of the welfare cost

Common margin. $\Delta \ln P_0 = \ln D/(1-\sigma_0)$ and $\Delta \ln Y = 0$, so $\Delta \ln W = -\ln D/(1-\sigma_0) < 0$.

Heterogeneous margins. Expand to first order in $u \equiv \Delta \ln P_g$: $\nu \approx 1 + (1-\sigma_0)u$ and $D \approx 1 + S_g(1-\sigma_0)u$, so $\Delta \ln P_0 \approx S_g u$ and

$$S_g^R - S_g \;\approx\; S_g(1-S_g)(1-\sigma_0)u, \qquad \Delta \ln Y \;\approx\; \frac{\kappa^R - \kappa}{1-\kappa} \;=\; S_g(1-S_g)(1-\sigma_0)u\,\frac{m_g - m_{-g}}{1-\kappa}.$$

Collecting terms in $\Delta \ln W = \Delta \ln Y - \Delta \ln P_0$ gives the formula.

The bracket is positive. If $m_g \le m_{-g}$, it exceeds one. Otherwise $(1-\sigma_0)(m_g - m_{-g}) \le 1 - m_{-g}$ and $(1-S_g)(1-m_{-g}) < 1-\kappa$, since the difference is $S_g(1-m_g) > 0$. The subtracted term is below one. ←

Proof of Proposition 2

Each firm's first-order condition is $p - c_i = b\,q_i$, so

$$q_L = \frac{(a - c_L) + n(c_f - c_L)}{(n+2)b}, \qquad q_f = \frac{(a - c_L) - 2(c_f - c_L)}{(n+2)b}, \qquad p = \frac{a + c_L + n c_f}{n+2}.$$

Differentiating total surplus, with $dq_L/dc_f = n/((n+2)b)$ and $dq_f/dc_f = -2/((n+2)b)$,

$$\frac{dTS}{dc_f} \;=\; \sum_i (p - c_i)\frac{dq_i}{dc_f} - n\,q_f \;=\; \frac{n}{n+2}\big[q_L - (n+4)\,q_f\big].$$

The bracket equals $\big[-(n+3)(a - c_L) + (3n+8)(c_f - c_L)\big]/((n+2)b)$, positive iff $(c_f - c_L)/(a - c_L) > (n+3)/(3n+8)$. The leader's share s_L is increasing in $(c_f - c_L)/(a - c_L)$; at the threshold it equals

$$s_L^*(n) \;=\; \frac{n+4}{2(n+2)}.$$

←

More on the data

Balanced store panel

	Stores	Retailers
Active in any year, 2012–2019	55,966	199
Active in every year	27,215	88
Reporting in all 32 quarters	26,966	67

- Stores present in every year keep the set of stores fixed before and after the event.
- The 67 uninterrupted retailers account for 99% of revenue.
- Zeros: a firm–county–quarter cell with zero revenue is kept if the firm sold in that county before and still sells elsewhere nationally. Zero cells are 5–33% of cells; PPML accommodates them.

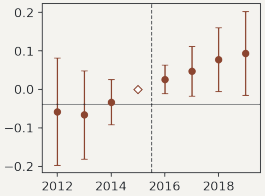
Share of zero-revenue cells

Product	Zero share
Nuts, oil-roasted	0.14
Nuts, dry-roasted	0.21
Nuts, raw	0.24
Dried fruit, true-dried	0.29
Dried fruit, fruit snacks	0.20
Peanut butter, creamy	0.19
Peanut butter, chunky	0.21
Chocolate, nut/PB	0.23
Chocolate, plain	0.33
Granola bars, w/ inclusions	0.13
Granola bars, plain	0.18
Granola cereal	0.17
Cereal	0.05
Seasoning	0.15
Crackers, flavored	0.15

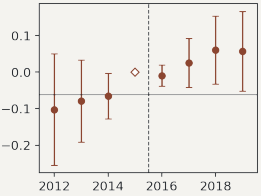
←

Remaining event studies: no pre-trends

Peanut butter, creamy



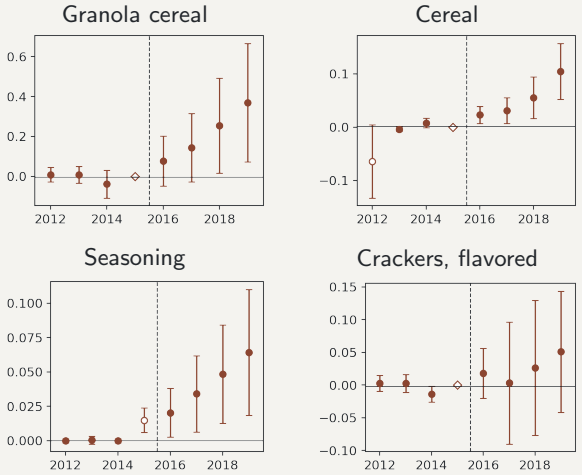
Peanut butter, chunky



Estimates of the elasticity of incidence by calendar year, with 95% confidence intervals. Compliance began in September 2016.

←

Remaining event studies: linear pre-trends



Elasticity of incidence by calendar year, net of the linear pre-trend, with 95% confidence intervals. Compliance began in September 2016.

←

Elastic outer nest ($\sigma_0 > 1$)

With $\sigma_0 > 1$, expenditure leaves industry g as its price index rises: $B < 0$. The most productive firms can still gain.

Theorem 3 (Gaining under an elastic outer nest)

Let $\sigma_0 > 1$ and $\sigma_g > \sigma_0$. Suppose incidence is regressive, with $\varepsilon(\psi_{\min}) = 0$. Under a common margin across industries, the most productive firm gains revenue and profit:

$$\Delta \ln r(\psi_{\min}) \;>\; (\sigma_g - \sigma_0)\, \Delta \ln P_g \;>\; 0.$$

Because $\Delta \ln r(\psi)$ is strictly decreasing in ψ, the winning firms form a top segment of the productivity distribution. With heterogeneous margins, the same conclusions hold if $\sigma_h \geq 2$ for every industry h. → Proof.

←

Pareto marginal costs (quantile form)

- Assume marginal costs are Pareto, with shape θ and scale $\psi_{\min}$:

$$\psi_\omega \;=\; G^{-1}(\omega) \;=\; \psi_{\min}\,(1-\omega)^{-1/\theta}, \qquad \omega \in (0,1).$$

- With a constant elasticity of incidence τ, incidence can be written as

$$1+\varepsilon(\psi) \;=\; \underbrace{\big(1+\varepsilon(\psi_{\min})\big)}_{\text{proportional}}\;\underbrace{\left(\frac{\psi}{\psi_{\min}}\right)^{\tau}}_{\text{regressive component}}.$$

- The elasticity of incidence τ changes the shape θ, and the proportional component $1+\varepsilon(\psi_{\min})$ changes the scale $\psi_{\min}$:

$$\psi^R_\omega \;=\; \big(G^R\big)^{-1}(\omega) \;=\; \big(1+\varepsilon(\psi_{\min})\big)\,\psi_{\min}\,(1-\omega)^{-(1+\tau)/\theta}.$$

←

Welfare with heterogeneous margins

With heterogeneous margins, welfare decreases to first order in $\Delta \ln P_g$:

Proposition (Welfare cost)

$$\Delta \ln W^{(1)} \;=\; -\,S_g\,\Delta \ln P_g \left[1-(1-\sigma_0)(1-S_g)\,\frac{m_g - m_{-g}}{1-\kappa}\right] \;<\; 0.$$

→ Proof.

- To first order, the loss is the industry's expenditure share times its price-index rise, adjusted by the bracket.
- The bracket captures the income response (Y rises if $m_g > m_{-g}$) and remains strictly positive.

←

Sensitivity to σ_0 and σ_g

	$\sigma_g = 3.9$			$\sigma_0 = 0.5$	
Product	$\sigma_0 = 0.5$	$\sigma_0 = 1$	$\sigma_0 = 2$	$\sigma_g = 3.1$	$\sigma_g = 5.1$
Nuts, oil-roasted	41 (21)	40 (18)	38 (12)	58 (22)	29 (20)
Nuts, dry-roasted	44 (22)	44 (19)	42 (13)	61 (23)	31 (21)
Dried fruit, true-dried	19 (20)	19 (18)	18 (12)	27 (21)	13 (20)
Peanut butter, creamy	26 (21)	26 (18)	25 (12)	36 (22)	18 (20)
Peanut butter, chunky	5 (21)	5 (19)	5 (13)	8 (23)	4 (21)
Chocolate, nut/PB	27 (17)	27 (15)	27 (10)	37 (18)	19 (16)
Granola bars, w/ inclusions	56 (19)	55 (16)	53 (11)	79 (20)	40 (18)
Granola cereal	12 (28)	12 (24)	12 (17)	17 (29)	9 (27)
Cereal	105 (16)	104 (14)	103 (10)	145 (17)	74 (16)
Seasoning	15 (19)	15 (16)	15 (11)	21 (20)	11 (18)
Crackers, flavored	11 (17)	11 (15)	11 (10)	15 (18)	8 (17)
Total	362	358	350	502	255

Welfare cost ($M/yr) and, in parentheses, the share of firms gaining (%). The first column is the baseline; $\sigma_g \in \{3.1, 5.1\}$ is the interquartile range in Hottman et al. (2016). The cost is insensitive to σ_0; the share of firms gaining falls as σ_0 rises.

←

Standard errors and confidence intervals

Product	$\hat{\beta}$ (SE)	$\hat{\theta}$ (SE)	$\Delta \ln P_g$ (%) (SE)	Welfare cost ($M/yr)	95% CI
Nuts, oil-roasted	0.456 (0.092)	0.89 (0.18)	10.3 (1.8)	41	[27, 56]
Nuts, dry-roasted	0.238 (0.077)	0.86 (0.15)	5.8 (1.7)	44	[18, 70]
Dried fruit, true-dried	0.292 (0.096)	0.80 (0.18)	7.1 (2.1)	19	[8, 31]
Peanut butter, creamy	0.103 (0.039)	0.74 (0.21)	2.7 (1.0)	26	[7, 45]
Peanut butter, chunky	0.097 (0.050)	0.79 (0.24)	2.5 (1.3)	5	[0, 11]
Chocolate, nut/PB	0.039 (0.028)	0.57 (0.10)	1.1 (0.8)	27	[−10, 64]
Granola bars, w/ inclusions	0.348 (0.188)	0.74 (0.20)	8.4 (4.0)	56	[2, 111]
Granola cereal	0.256 (0.120)	1.26 (0.38)	5.7 (2.5)	12	[2, 22]
Cereal	0.064 (0.019)	0.54 (0.11)	1.8 (0.5)	105	[45, 164]
Seasoning	0.049 (0.018)	0.64 (0.11)	1.4 (0.5)	15	[4, 26]
Crackers, flavored	0.027 (0.048)	0.57 (0.12)	0.8 (1.4)	11	[−26, 47]
Total				362	[258, 466]

The standard errors of $\Delta \ln P_g$ and the welfare cost propagate the standard error of $\hat{\beta}$ by the delta method, holding θ, $\sigma_g = 3.9$, $\sigma_0 = 0.5$, and S_g fixed. The total's interval adds variances across products.

←

Cost-function space: isolation and the reduction

Definition (isolated cost-minimization)

Firm i takes factor prices r as given and r is unaffected by the choices of any **individual** firm.

Proposition (reduction)

Under isolation, the firm's problem separates: compute $C_i(q_i)$ from own output and given factor prices; then maximize $p_i q_i - C_i(q_i)$. The industry equilibrium is characterized entirely by $\{C_i\}$ and demand.

- Consistent with r determined in equilibrium; what is ruled out is factor-market **power**.
- Constant returns + common overhead give the affine form $C_i(q) = c_i q + F$ with $c_i = e(r)/\varphi_i$ — a common **level** (neutral) and a cross-firm **pattern** (where

GE rung 1: endogenous income, fixed mass

Shares are homogeneous of degree zero in Y: the who-wins characterization carries over **verbatim for shares**. For levels, under CES:

$$\Delta \ln r_\omega \;=\; (\sigma - 1)\big[\Delta \ln P - \Delta \ln \psi_\omega\big] + \Delta \ln Y.$$

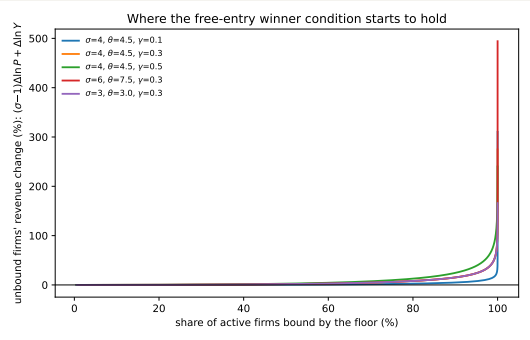
Proposition (regressive incidence in GE)

Cost share smallest at the top $\Rightarrow$ (i) top **shares** gain exactly, regardless of income; (ii) top **levels** change by $(\sigma - 1)\big(\mathbb{E}_s[\varepsilon] - \varepsilon_0\big) + \Delta \ln Y$, first term strictly positive — gains whenever income does not contract by more than the relative-price gain.

First-order aggregator response = the share-weighted average burden (CES, fixed mass). For one grocery category, Y barely moves.

GE rung 2: entry, and the calibration

With sunk entry (Melitz), adding-up renormalizes over an endogenous set: the variety margin can push the aggregator **down** for small deformations — the one rung where the guarantees lapse.



Calibration (rent-sharing free-entry economy, conventional grids): **the winner condition holds from the first binding floor in all 44 of 44 admissible cells.** At

References I

Andreyeva, T., M. W. Long, and K. D. Brownell (2010): "The Impact of Food Prices on Consumption: A Systematic Review of Research on the Price Elasticity of Demand for Food," *American Journal of Public Health*, 100, 216–222.

Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen (2020): "The Fall of the Labor Share and the Rise of Superstar Firms," *Quarterly Journal of Economics*, 135, 645–709.

Covarrubias, M., G. Gutiérrez, and T. Philippon (2020): "From Good to Bad Concentration? US Industries over the Past 30 Years," *NBER Macroeconomics Annual*, 34, 1–46.

De Loecker, J., J. Eeckhout, and G. Unger (2020): "The Rise of Market Power and the Macroeconomic Implications," *Quarterly Journal of Economics*, 135, 561–644.

Dobkin, C., A. Finkelstein, R. Kluender, and M. J. Notowidigdo (2018): "The Economic Consequences of Hospital Admissions," *American Economic Review*, 108, 308–352.

Edmond, C., V. Midrigan, and D. Y. Xu (2023): "How Costly Are Markups?" *Journal of Political Economy*, 131, 1619–1675.

Fowlie, M., M. Reguant, and S. P. Ryan (2016): "Market-Based Emissions Regulation and Industry Dynamics," *Journal of Political Economy*, 124, 249–302.

References II

Freyaldenhoven, S., C. Hansen, J. Pérez Pérez, and J. M. Shapiro (2026): "Visualization, Identification, and Estimation in the Linear Panel Event-Study Design," in *Advances in Economics and Econometrics: Twelfth World Congress*, ed. by V. Chernozhukov, J. Hörner, E. La Ferrara, and I. Werning, Cambridge University Press, vol. 2, 225–268.

Gabaix, X. (2011): "The Granular Origins of Aggregate Fluctuations," *Econometrica*, 79, 733–772.

Hill, B. M. (1975): "A Simple General Approach to Inference about the Tail of a Distribution," *The Annals of Statistics*, 3, 1163–1174.

Hottman, C. J., S. J. Redding, and D. E. Weinstein (2016): "Quantifying the Sources of Firm Heterogeneity," *Quarterly Journal of Economics*, 131, 1291–1364.

Kwon, S. Y., Y. Ma, and K. Zimmermann (2024): "100 Years of Rising Corporate Concentration," *American Economic Review*, 114, 2111–2140.

Macedoni, L. and A. Weinberger (2026): "Lobbying for Regulations: When Big Business Says Yes," Working Paper.

Matsuyama, K. and P. Ushchev (2017): "Three Alternative Classes of Flexible Homothetic Demand Systems," SSRN Working Paper.

References III

Rebeyrol, V. (2023): "Protection Without Discrimination," *Economic Journal*, 133, 2807–2827.

Salop, S. C. and D. T. Scheffman (1983): "Raising Rivals' Costs," *American Economic Review*, 73, 267–271.

Santos Silva, J. M. C. and S. Tenreyro (2006): "The Log of Gravity," *Review of Economics and Statistics*, 88, 641–658.

Williamson, O. E. (1968): "Wage Rates as a Barrier to Entry: The Pennington Case in Perspective," *Quarterly Journal of Economics*, 82, 85–116.